

Preliminary analysis of newly recovered Apollo 17 seismic data

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ABSTRACT

Using a novel pattern recognition algorithm, up to 50,000 new seismic events have been detected on the Moon over an 8-month period from newly rediscovered Apollo 17 seismic data. These events occur periodically, with a sharp double-peak at sunrise and a broad single-peak at sunset. Due to the proximity of the events to the Apollo 17 site, and the diurnal timing of the peaks, we interpret these events to be thermally-driven – caused by rapid heating and cooling of materials at the surface of the Moon as the sun passes overhead. Locating the source of these detected events is ongoing, and could be valuable for identifying which surface features are the source of the events. This will lead to a better understanding of the conditions under which they occur. This body of work is particularly relevant to studies of lunar regolith breakdown, because thermal seismic events at the surface of the Moon are an expression of an active degradational process.

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Introduction

During the Apollo 17 mission to the Moon, astronauts deployed a small network of four geophones on the lunar surface for an active-source seismic experiment called the Lunar Seismic Profiling Experiment (LSPE) [1]. The primary experiment was completed shortly after the astronauts departed, but the geophones were later operated in “listening mode,” where they could record natural seismic signals. Listening mode segments from 1974 totaling about one month have previously been analyzed, and detected what appeared to be thermal moonquakes [2]. In this study, we examine a longer 8.3 month segment that is generally continuous between August 1976 and April 1977, and has not yet been systematically searched for seismic events.

Method

We use a Hidden Markov Model (HMM) based pattern recognition technique to jointly detect and categorize seismic events [3]. HMM's have previously detected seismic events on Apollo 16 passive seismic data [4], and we follow the same basic approach for Apollo 17 geophone data. In this study, three events with unique waveform characteristics were used as templates (Fig. 1). Continuous waveform data from Geophone 1 were sequentially analyzed,

and every 30 s classified either as one of the three template types or as background noise. The template detections for each event type form the basis of a seismic event catalog unique to Apollo 17.

Results and discussion

A total of 49,635 detections were made. Of these, 2249 (4.5%) are termed “impulsive”, 13,355 (26.9%) are “intermediate”, and 34,031 (68.5%) are “emergent.” These terms describe the seismic event risetime, where impulsive events reach their maximum energy quickly, emergent events take a longer time, and intermediate events are somewhere in between (Fig. 1). We interpret all events as local to the Apollo 17 site, as known distant moonquakes from the lunar seismic catalog [5] could not be identified. Impulsive events reach a maximum at sunrise while emergent events reach a maximum at sunset (Fig. 2). One possibility is that impulsive events originate from the Apollo 17 lunar lander, while emergent events are from the regional soil and rocks. The lander is metallic and therefore heats up quicker than the lunar regolith, with event numbers peaking sharply soon after sunrise, whereas thermal events in the regolith peak more gradually and closer to sunset. Each sunrise peak in Fig. 2a is actually two peaks separated by about 20 h in time (Fig. 2d). We are currently investigating if the two peaks at sunrise could be caused by the Sun's progression over the angled shape of the lander, the passing of shadows from adja-

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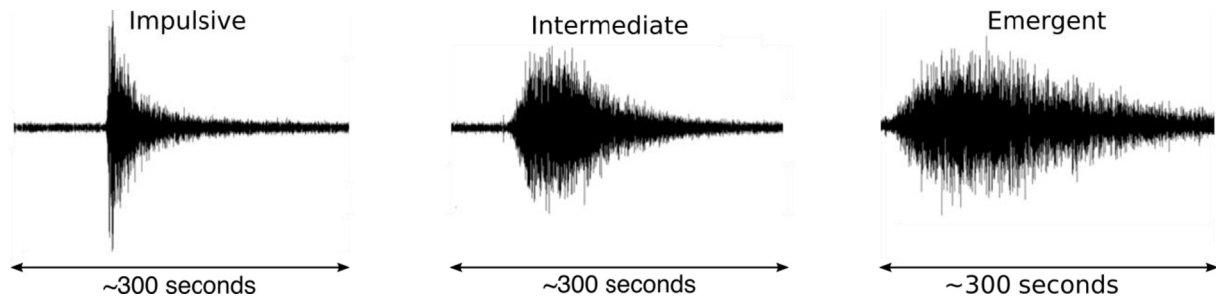


Fig. 1. Seismic event templates cut from Apollo 17 seismic data and used for HMM pattern recognition.

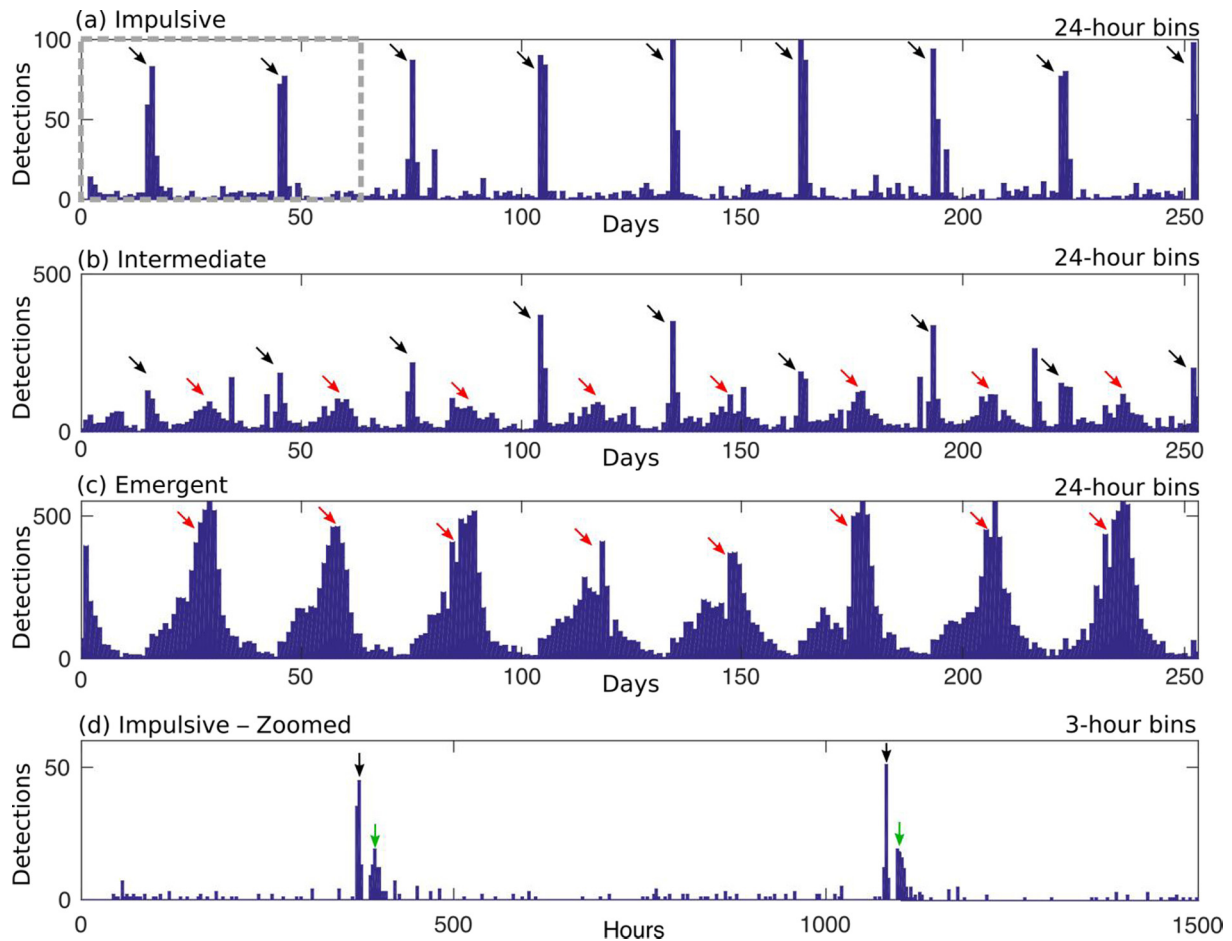


Fig. 2. Histograms of impulsive, intermediate, and emergent seismic events detected on 8.3 months of Apollo 17 seismic data. Black arrows = sunrise peak; red arrows = sunset peak; green arrows = second sunrise peak. Gray dashed region in (a) is zoomed in (d). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

cent mountains (North Massif and South Massif), or might represent another local source such as craters or boulders.

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